```csharp
using System;

namespace StudentApp
{
    // A class is a blueprint for creating objects.
    // It defines properties (fields) and behaviors (methods).
    class Student
    {
        // Fields (variables inside the class)
        public string name;
        public int age;
        public double grade;

        // Constant (same value for all objects)
        public const string universityName = "Yarmouk University";

        // Static member (shared among all objects)
        public static int studentCount = 0;

        // Constructor (called when object is created)
        public Student(string name, int age, double grade)
        {
            this.name = name;
            this.age = age;
            this.grade = grade;

            studentCount++; // Increase count when a new student is created
        }

        // Method to display student information
        public void DisplayInfo()
        {
            Console.WriteLine("Name: " + name);
            Console.WriteLine("Age: " + age);
            Console.WriteLine("Grade: " + grade);
            Console.WriteLine("University: " + universityName);
        }

        // Method to update grade
        public void UpdateGrade(double newGrade)
        {
            grade = newGrade;
        }

        // Method to check if student passed
        public bool IsPassed()
        {
            return grade >= 50;
        }
    }

    class Program
    {
        static void Main(string[] args)
        {
            // An object is an instance of a class.
            // Example: s1, s2, s3 are objects created from Student class.
```

```csharp
Student s1 = new Student("Anwar", 23, 75.5);
Student s2 = new Student("Mohammad", 19, 45.0);
Student s3 = new Student("Hassan", 21, 60.0);

// Display info and pass status for each student
Student[] students = { s1, s2, s3 };

foreach (Student s in students)
{
    s.DisplayInfo();

    if (s.IsPassed())
        Console.WriteLine("Status: Passed");
    else
        Console.WriteLine("Status: Failed");

    Console.WriteLine("----------------------");
}

// Update grade example
s2.UpdateGrade(55);

Console.WriteLine("After updating Lina's grade:");
s2.DisplayInfo();
Console.WriteLine("Status: " + (s2.IsPassed() ? "Passed" : "Failed"));

Console.WriteLine("----------------------");

// Display total number of students
Console.WriteLine("Total Students Created: " + Student.studentCount);

    // 1. Even or Odd
    CheckEvenOdd(7);

    // 2. Second smallest
    int[] arr1 = { 4, -3, 7, 2, 0 };
    SecondSmallest(arr1);

    // 3. Factorial
    Factorial(5);

    // 5. Largest number in array
    int[] arr2 = { 3, 1, 4, 1, 5, 9 };
    FindLargest(arr2);

    // 6. Number pattern
    NumberPattern(5);

    // 7. Star pyramid (optional)
    StarPyramid(4);

    // 8. Sum of even & odd (optional)
    int[] arr3 = { 1, 2, 3, 4, 5, 6 };
    SumEvenOdd(arr3);

    // 9. Common elements (optional)
    int[] a = { 1, 2, 3, 4 };
    int[] b = { 3, 4, 5, 6 };
```

```
    CommonElements(a, b);
}

// 1. Even or Odd
static void CheckEvenOdd(int number)
{
    if (number % 2 == 0)
        Console.WriteLine($"The number {number} is even.");
    else
        Console.WriteLine($"The number {number} is odd.");
}

// 2. Second smallest number
static void SecondSmallest(int[] arr)
{
    Array.Sort(arr);
    Console.WriteLine("Second smallest: " + arr[1]);
}

// 3. Factorial using for loop
static void Factorial(int n)
{
    int result = 1;

    for (int i = 1; i <= n; i++)
    {
        result *= i;
    }

    Console.WriteLine($"Factorial of {n} = {result}");
}

// 5. Find largest using foreach
static void FindLargest(int[] arr)
{
    int max = arr[0];

    foreach (int num in arr)
    {
        if (num > max)
            max = num;
    }

    Console.WriteLine("Largest number: " + max);
}

// 6. Number pattern
static void NumberPattern(int n)
{
    int num = 1;

    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            Console.Write(num + " ");
            num++;
        }
```

```
            Console.WriteLine();
        }
    }

    // 7. Star pyramid (optional)
    static void StarPyramid(int n)
    {
        for (int i = 1; i <= n; i++)
        {
            // spaces
            for (int j = 1; j <= n - i; j++)
                Console.Write(" ");

            // stars
            for (int k = 1; k <= (2 * i - 1); k++)
                Console.Write("*");

            Console.WriteLine();
        }
    }

    // 8. Sum of even and odd numbers
    static void SumEvenOdd(int[] arr)
    {
        int evenSum = 0, oddSum = 0;

        foreach (int num in arr)
        {
            if (num % 2 == 0)
                evenSum += num;
            else
                oddSum += num;
        }

        Console.WriteLine("Sum of Even Numbers: " + evenSum);
        Console.WriteLine("Sum of Odd Numbers: " + oddSum);
    }

    // 9. Common elements in two arrays
    static void CommonElements(int[] a, int[] b)
    {
        Console.Write("Common elements: ");

        foreach (int x in a)
        {
            foreach (int y in b)
            {
                if (x == y)
                {
                    Console.Write(x + " ");
                    break;
                }
            }
        }

        Console.WriteLine();
    }
}
```

```
}
```

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```
Name: Anwar
Age: 23
Grade: 75.5
University: Yarmouk University
Status: Passed
----------------------
Name: Mohammad
Age: 19
Grade: 45
University: Yarmouk University
Status: Failed
----------------------
Name: Hassan
Age: 21
Grade: 60
University: Yarmouk University
Status: Passed
----------------------
After updating Lina's grade:
Name: Mohammad
Age: 19
Grade: 55
University: Yarmouk University
Status: Passed
----------------------
Total Students Created: 3
The number 7 is odd.
Second smallest: 0
Factorial of 5 = 120
Largest number: 9
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
   *
  ***
 *****
*******
Sum of Even Numbers: 12
Sum of Odd Numbers: 9
```

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```
----------------------
Name: Mohammad
Age: 19
Grade: 45
University: Yarmouk University
Status: Failed
----------------------
Name: Hassan
Age: 21
Grade: 60
University: Yarmouk University
Status: Passed
----------------------
After updating Lina's grade:
Name: Mohammad
Age: 19
Grade: 55
University: Yarmouk University
Status: Passed
----------------------
Total Students Created: 3
The number 7 is odd.
Second smallest: 0
Factorial of 5 = 120
Largest number: 9
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
   *
  ***
 *****
*******
Sum of Even Numbers: 12
Sum of Odd Numbers: 9
Common elements: 3 4

C:\Users\user\OneDrive\Desktop\day4\StudentApp\StudentApp\bin\Debug\StudentApp.exe (process 7204) exited with code 0 (0x0).
Press any key to close this window . . .
```